

GOVERNANCE · RESILIENCE

Inventre

Disaster Recovery & Business Continuity

What we do when the server dies, the data is damaged, a partner is down or the people are unavailable: the targets we commit to, who does what, the exact steps, and how we prove it works before we need it.



DOCUMENT	DATE	STATUS	OWNER
DR-BCP v1.3	24 September 2026	Baseline; first restore drill recorded	Inventre Engineering

Technical procedures reference the runbooks in the repository (docs/runbooks/) and the recovery kit stored beside the backups. Names in the contact table are placeholders where a role has not yet been assigned; filling them is the first action.

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Revision history

Version	Date	Author	Change
1.0	2026-09-23	Engineering	First baseline after the backup pipeline and recovery kit went live; targets proposed, playbooks written, drill schedule set.
1.1	2026-09-24	Engineering	Point-in-time restore drill run on the production host into a throwaway container (D-02); one-command maintenance switch (D-08); restore script gains a volume override; new finding D-11 (nginx bypass token readable).
1.2	2026-09-24	Engineering	D-11 closed: the maintenance bypass token moved out of the world-readable site file into a root-only include.
1.3	2026-09-24	Engineering	Clean-VM rebuild drill staged on the production host (D-01); SharePoint retention runbook (D-06); standby recommendation (D-07); offline-order SOP and paper form (D-09); partner-contact checklist (D-10); new finding D-12.

01 Summary and commitments

Inventre runs on one server. Until this month, losing it would have meant losing up to 6 hours of orders and the entire code history since June. As of 23 September 2026 the database is streamed off the server continuously, the code is pushed to GitHub on every commit, and everything needed to rebuild lives, encrypted, in the company's Microsoft 365 alongside a one-command recovery kit. What remains is to agree the targets below, name the people, and rehearse.

Commitments proposed for approval

Measure	Target	What it means
RPO (Recovery Point Objective): most data we may lose	5 minutes	After a total loss, orders placed more than 5 minutes before the failure are recovered. Achieved by continuous transaction-log streaming plus a 5-minute off-host sync.
RTO (Recovery Time Objective): longest acceptable downtime	4 hours business hours Next morning overnight	The technical rebuild is 60 to 90 minutes; the rest is people, DNS and partner steps. Overnight incidents (outside 07:00 to 21:00 IST) are recovered by 09:00.
Availability during the ordering season (May to August)	99.5 % in business hours	About 1 hour of unplanned downtime per month in business hours.
Maximum tolerable outage before the business is materially hurt	2 business days	Beyond this, schools escalate, parents lose trust and delivery slots are missed. Every playbook is designed to finish far inside it.
Drill frequency	Quarterly restore drill, annual full rebuild	A backup that has never been restored is a hope, not a backup.

CURRENT MEASURED POSITION

RPO: about 5 minutes (verified: transaction-log segments and the 5-minute mirror are live). RTO for a full rebuild (Scenario A): **not yet measured**; the rebuild script exists and is syntax-checked but has not been run on a clean machine. The point-in-time restore (Scenario B) was drilled on 2026-09-24 into a throwaway container on the production host; the measured time is in the drill record (Section 9). The first full rebuild drill turns the 90-minute estimate into a number.

02 What we protect and what it depends on

2.1 Business functions, in order of importance

Function	Criticality	Why, and what stops if it fails
Parents ordering and paying	Critical	Revenue. In season, hundreds of orders a day; a down storefront during a school's ordering window is visible to the school immediately.
Order flow to fulfilment (Audit ERP)	Critical	Paid orders that do not reach the warehouse are not packed. The queue buffers; a day's delay is recoverable, a week's is not.
Payment reconciliation (CCAvenue)	Important	Money is safe at the gateway; unreconciled orders show as unpaid until healed. Recoverable from settlement files.
Admin console (operations)	Important	Staff cannot change catalogue, students or orders. Tolerable for hours, not days.
Fee ledger console	Standard	Management reporting from MyClassBoard. Source of truth is MyClassBoard; a day's outage is an inconvenience.
Exchange, missing-item and concern intake	Standard	Parents can phone the call centre instead; requests are back-filled.

2.2 Dependencies

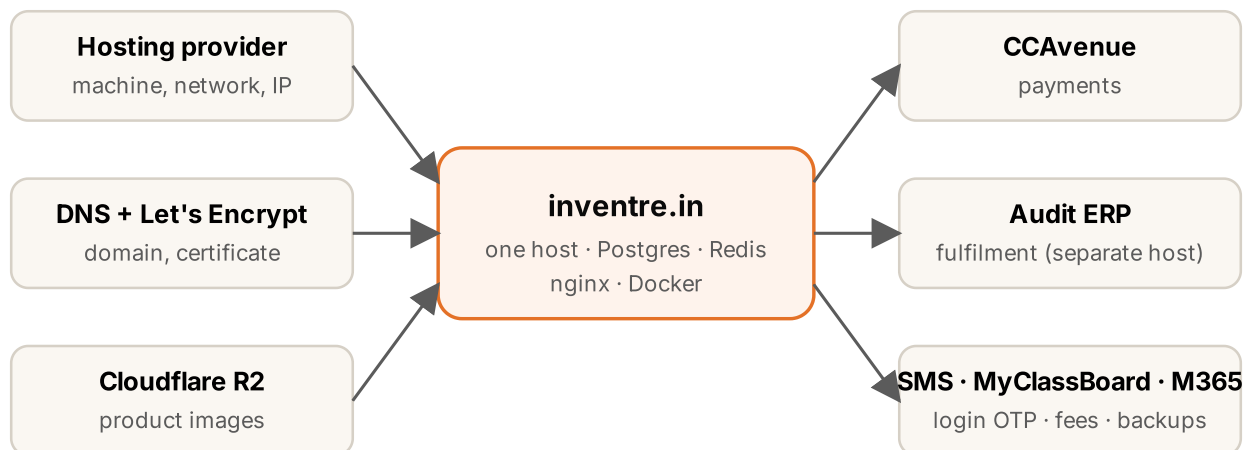


Figure 1 — Everything the storefront needs to take an order. The single host in the middle is the only component without redundancy.

Dependency	If it fails	Fallback
The host	Everything stops	Rebuild on a new machine from the backups (Scenario A)
Hosting provider network or IP	Everything stops, data intact	Wait or rebuild elsewhere; DNS change
DNS or certificate	Browsers refuse or cannot find the site	Registrar login; certbot renews automatically, manual renew if it did not
Cloudflare R2	Product images missing; ordering still works	Wait; images are also in the 6-hourly snapshot's media volume
CCAvenue	No online payment	Offline / pay-at-school orders can be recorded by staff; reconcile later
Audit ERP	Orders queue up; status stops updating	Queue holds 5 attempts with backoff and a dead-letter table; replay from admin when back
SMS gateway	No OTP login	Password login exists; staff can toggle the OTP bypass for a supervised window
MyClassBoard	Fee console shows last night's data	Nightly import retries; no storefront impact

Dependency	If it fails	Fallback
Microsoft 365	No alert e-mails; backup mirror pauses (local copies continue)	Sync resumes on its own; logs on the host
GitHub	Code push pauses	Bundles in the mirror carry the code

03 Targets: how much loss and downtime we accept

Function	RPO	RTO (business hours)	How the target is met
Storefront ordering and payment	5 min	4 h	Transaction-log streaming + 5-minute mirror; recovery kit rebuild; DNS switch
Order flow to fulfilment	5 min	4 h (same rebuild)	Outbound queue is in the database, so it is restored with it; the ERP re-drains on the next tick
Payment reconciliation	0 (source is CCAvenue)	24 h	Settlement files and the Status API rebuild any missing payment state
Admin console	5 min	4 h	Same application
Fee ledger	24 h (nightly import)	Next morning	Re-import from MyClassBoard; nothing unique is stored here
Code and configuration	seconds (commit push) / 5 min (uncommitted, env)	minutes	GitHub + bundles + encrypted env copy

WHAT THE TARGETS DO NOT COVER

A failure at CCAvenue, the Audit ERP or the schools' own systems is outside Inventre's control; the fallbacks in Section 2.2 limit the damage but the recovery clock belongs to the partner. A deliberate attack that also destroys the Microsoft 365 backups would need the tenant's own recycle-bin and retention protections; enabling retention locks on the backup library is action D-06.

04 People: roles and contacts

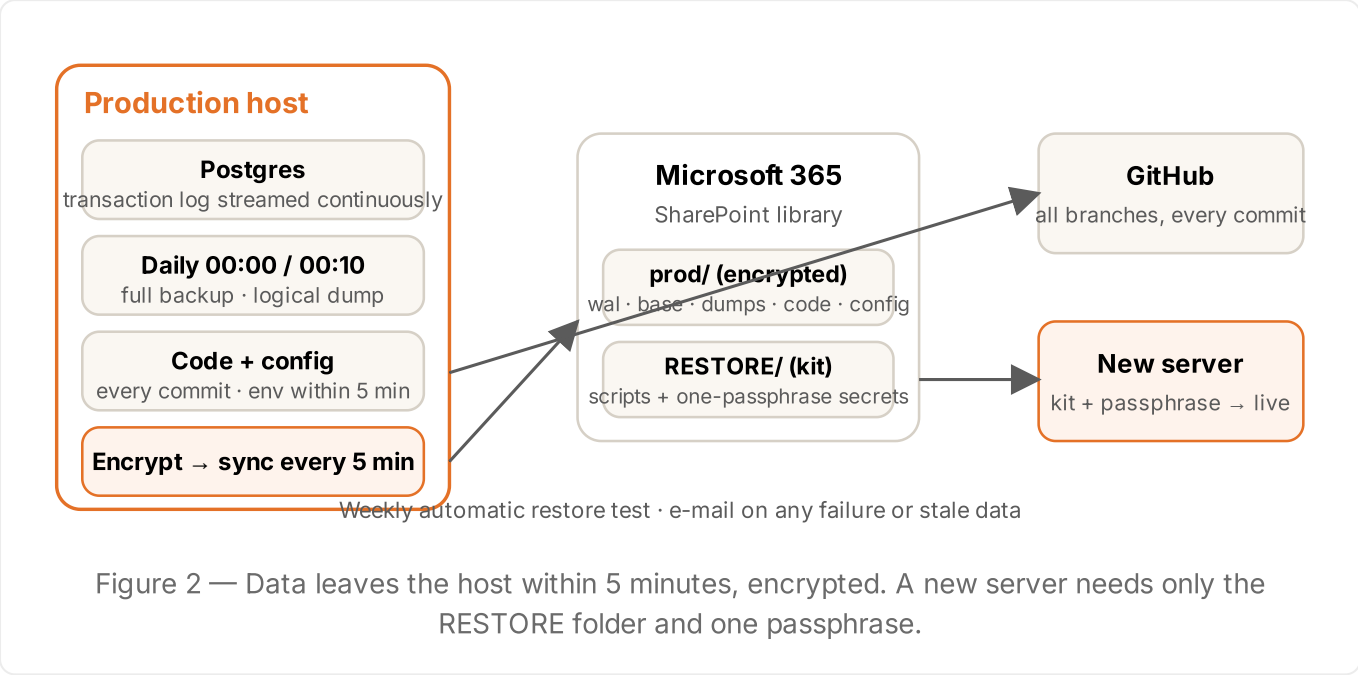
Incidents fail on people, not scripts. Three roles, each with a deputy. Fill the blanks and keep this table in the password manager as well, since the server may be the thing that is down.

Role	Primary	Deputy	Decides / does
Incident owner	Company owner — sunith@inventre.in	<i>to be named</i>	Declares the incident, approves the maintenance page, approves a rebuild or a point-in-time rollback, talks to schools, authorises spend (new server).
Technical lead	Inventre Engineering (this role is currently filled through the AI-assisted engineering sessions with the owner)	<i>to be named</i>	Runs the playbook, takes snapshots before changing anything, keeps the timeline, restores service.
Communications lead	Operations lead — <i>to be named</i>	<i>to be named</i>	Tells schools, parents and the call centre what is happening and when to expect service; uses the templates in Section 7.
Alert mailbox	it@inventre.in (all automated alerts), plus sudheer@, sriram@ for stranded-order alerts		Someone must read it; set a phone notification for it.
Escalation partners	Audit ERP team (fulfilment host, 217.216.58.218) · CCAvenue merchant support (merchant id on file) · hosting provider support · Microsoft 365 admin · domain registrar		

What each person must be able to reach without the server

- The password manager entry **Inventre recovery kit passphrase**, and the encryption keys and snapshot passphrase stored with it.
- SharePoint: *Vendor Management Files AUDIT > Documents > Inventre Backups* (the **RESTORE** folder is readable; **prod** is encrypted).
- GitHub: **Sunith0801/Inventre-Website**.
- Hosting provider console, domain registrar, Cloudflare account, Microsoft 365 admin centre, CCAvenue merchant panel.
- This document and the runbooks (also in GitHub under **docs/**).

05 The backup system, as built



Layer	Detail	Retention
Transaction-log streaming	<code>pg_receivewal</code> as a systemd service; any-second recovery together with the daily full backup	31 days
Full physical backup	00:00 IST, <code>pg_basebackup</code>	30 days
Logical dump	00:10 IST, <code>pg_dump</code> ; the simple restore path	14 daily, monthly for 12 months
Snapshot	Every 6 h: dump + encrypted configuration + media volume	7 days local, 30 remote
Configuration	<code>.env.deploy</code> re-encrypted within 5 minutes of a change	latest
Code	Bundle of all branches on every commit, after every deploy and 00:20; GitHub push on every commit; uncommitted edits every 5 min	14 bundles; GitHub forever
Off-host mirror	<code>rclone</code> every 5 min, client-side encrypted (names and contents), to SharePoint	mirrors local retention
Recovery kit	<code>RESTORE/</code> : bootstrap script, restore engine, README, all secrets under one passphrase; republished every sync	latest

Layer	Detail	Retention
Proof	Sunday 03:30 restore of the newest dump into a throwaway database with a pass/fail e-mail; alerts when the mirror fails or the stream stops	—

06 Scenarios and playbooks

Each playbook starts the same way: **declare, snapshot, then act**. Nothing is rebooted, deleted or redeployed until the technical lead has written the first timeline line and, where the host is alive, taken a snapshot.

Scenario A — The server is lost (hardware, provider, ransomware, fire)

Step	Action	Who	Time
A1	Declare the incident; put the maintenance notice up if any part of the old host still answers; send the first message (Section 7, template 1).	Owner, comms	0:00–0:15
A2	Order a new Ubuntu server (4 vCPU, 16 GB, 200 GB) from the provider; note its IP.	Owner	0:15–0:30
A3	Download the RESTORE folder from SharePoint, copy it to the new server, run bash bootstrap-new-server.sh , enter the recovery-kit passphrase. It fetches, restores code and configuration, replays the database to the last received segment and starts the app.	Technical lead	0:30–1:45
A4	Check the counts the script prints (tables, orders, students, newest order time). If the newest order is older than expected, stop and investigate before opening to parents.	Technical lead	1:45–1:55
A5	nginx + certificate, DNS A records to the new IP, cron files, WAL streaming service, mail relay, CCAvenue SFTP user. Rename the SharePoint prod folder to prod-old-<date> before the first sync from the new host.	Technical lead	1:55–2:30
A6	Verify: health endpoint, security headers, a test login, a test order to the CCAvenue test page, the ERP drain log shows sent . Tell CCAvenue and the ERP team the new outbound IP if they allow-list.	Technical lead	2:30–3:00
A7	Send the all-clear (template 3). Within two weeks: post-incident review, this document updated.	Comms, owner	3:00

RPO: about 5 minutes. RTO estimate: 3 hours including people time; technical part 60 to 90 minutes.

Reference: [docs/runbooks/restore-from-m365.md](#) .

Scenario B — Bad data: a faulty deploy, a wrong bulk import, a deleted table

Step	Action	Who	Time
B1	Stop the bleeding: roll the application back to the previous image tag (<code>docker tag ... rollback-*</code> , redeploy, 1 minute) if a deploy caused it; disable the offending cron or import if a job caused it.	Technical lead	0:00–0:10
B2	Decide the scope with the owner: a few rows (fix forward with a script), a table (restore that table from the last dump into a scratch database and copy back), or everything since a time (point-in-time restore).	Owner + technical lead	0:10–0:30
B3	For point-in-time: the maintenance page goes up (orders placed during the window would be lost), the technical lead runs <code>restore-from-m365.sh restore-pitr "YYYY-MM-DD HH:MM"</code> on the <i>same</i> host into a fresh data volume, verifies counts, then switches the compose stack to it. Orders placed between the target time and the restore are re-entered from CCAvenue's records.	Technical lead	0:30–1:30
B4	Verify, reopen, all-clear; record what was lost and re-entered.	Comms	1:30

Drill: `scripts/backup/pitr-drill.sh ["YYYY-MM-DD HH:MM:SS"]` rehearses step B3 without touching the live stack. It restores from the local archive (`/root/backups-archive` , which holds the WAL segments the replay needs) into a throwaway container and data volume on the production host (`PG_VOLUME` override), prints the counts, then removes them. First run 2026-09-24 (Section 9). The dev database cannot be used for this: it has no WAL archive.

Scenario C — Security compromise (repeat of June 2026)

Step	Action	Who
C1	Do not wipe. Snapshot first (<code>scripts/snapshot-and-rotate.sh</code>), copy logs off-host, keep the timeline. Maintenance page up.	Technical lead
C2	Contain: rotate every secret (<code>.env.deploy</code> values, database password, recovery-kit passphrase, GitHub key, Microsoft app secret), revoke all sessions by rotating the JWT secret without the previous-key window, block the attacker's source at the firewall.	Technical lead
C3	Assess personal-data exposure (Data Protection document, Section 11) and start the 72-hour notification clock if data may have been accessed.	Owner

Step	Action	Who
C4	Rebuild rather than clean: treat it as Scenario A onto a fresh server from a backup taken <i>before</i> the intrusion (point-in-time restore to that moment if the intrusion touched data). Do not restore code from the compromised host's working tree; use GitHub or a bundle from before.	Technical lead
C5	Post-incident review, credentials of every partner re-issued, this document updated.	Owner

Scenario D — A partner is down

Partner	Symptom	Response
CCAvenue	Payments fail or callbacks stop; stuck "pending" orders	Do nothing destructive. The 5-minute reconcile heals when the gateway returns. If prolonged, staff record offline orders for urgent cases; message parents (template 4).
Audit ERP	Drain log shows failures; stranded-order alert e-mail	Queue retries 5 times with backoff, then dead-letters. When the ERP is back: admin ERP bridge page → replay dead letters; run erp-reconcile by hand.
SMS gateway ("insufficient balance" or outage)	Parents cannot log in	Recharge or wait; meanwhile promote password login; a supervisor may enable the OTP bypass for a short, logged window.
MyClassBoard	Nightly import fails	Fee console shows yesterday; retry next night; no parent impact.
Microsoft 365	Backup sync alert; no e-mails	Local copies continue; sync resumes automatically. If longer than a day, copy the archive to a second location by hand.

Scenario E — Key people unavailable

The deputy in Section 4 takes the role. Everything a deputy needs is in the password manager and SharePoint, not on any one laptop. If the technical lead is unavailable, the recovery kit's README is written so that any competent engineer, or an AI assistant given the folder and the passphrase, can complete Scenario A.

07 Communication during an incident

Audience	Channel	When	Owner
Parents	Maintenance page on inventre.in (nginx 503 gate, already built); SMS only if outage exceeds 4 hours in season	Immediately; then every 2 hours	Comms lead (technical lead flips the page)
Schools	Phone call to the coordinator, then e-mail	Within 30 minutes if their ordering window is affected	Comms lead
Call centre (Audit ERP team)	Group chat	Immediately, with the parent-facing line to use	Comms lead
Partners (CCAvenue, ERP team)	Support ticket / call	When their allow-lists or credentials change	Technical lead

Templates (adapt, do not improvise)

- Initial:** "Inventre's online store is temporarily unavailable. Your orders and payments are safe. We are restoring service and expect to be back by [time]. Payments already made will show correctly once we are back."
- Update:** "Update [time]: restoration is under way; the store will reopen by [new time]. If you paid and do not see your order after we reopen, it will appear within an hour; no action is needed."
- All-clear:** "The store is back online. Orders placed before [time] are intact. If you paid between [time] and [time] and your order is missing, reply to this message or call the school coordinator; we will complete it for you."
- Payment partner outage:** "Online payment is currently unavailable due to an issue at our payment provider. You can complete your order later; your cart is saved. Urgent orders can be placed through the school coordinator."

MAINTENANCE PAGE

Already built: nginx serves a branded page with a 503 status when `$maint_on` is 1; admin and API paths stay live; staff bypass with a token. Copy the current page to nginx with `/root/maintenance/sync.sh`. One command: `bash scripts/maintenance.sh on|off|status` (backs up the nginx site file, edits the map, tests, reloads, restores on failure). Runbook `docs/runbooks/maintenance-page.md`; exact nginx steps in the HLD (maintenance page and nginx 503 gate).

08 Degraded operation: keeping the business going

Situation	How the business continues while systems are restored
Storefront down in an ordering window	School coordinator collects orders on paper or a shared sheet (child name, enrolment, items, sizes); staff key them in as offline orders when the store is back (the manual offline-order script exists) and reconcile payment separately.
Payments down	Accept pay-at-school or bank transfer for urgent orders; record as offline payments; the settlement reconcile heals gateway orders later.
Fulfilment link down	Warehouse works from the admin console's order list or an Excel export until the queue drains.
Login (OTP) down	Password login; supervised OTP bypass; call centre places orders on behalf of parents using view-as-parent is read-only, so orders are keyed as offline orders instead.
Fee console down	Fee desk works directly in MyClassBoard; Inventre's console is a reporting layer.
Engineering unavailable	Deputy follows the recovery kit README; the hosting provider's console can reboot the machine; the maintenance page can be toggled by anyone with SSH (<code>bash scripts/maintenance.sh on off</code>).

09 Testing and drills

Test	Frequency	What it proves	Status
Automatic restore of the newest dump	Weekly, Sunday 03:30	The dump is restorable; table and row counts are sane; e-mail to it@inventre.in	Scheduled first run 2026-09-27
Mirror integrity	Continuous	Sync succeeded; transaction log is fresh; e-mail on failure or staleness	Live
Full rebuild drill (Scenario A on a throwaway server)	Quarterly; first one now	The recovery kit works end to end; measures the real RTO; finds gaps in this document	Not yet done D-01
Point-in-time restore drill (Scenario B, throwaway container on the production host)	Quarterly	Replay to a chosen minute works; the team knows the command	Done 2026-09-24 — 45 s from the base backup + 12 WAL segments to a promoted database (118 tables, 31,683 orders, 24,213 students). The first attempt failed on the in-progress WAL segment; fixed in both restore scripts
Tabletop: compromise + data-breach notification	Annual	People know the order of actions and the 72-hour clock	Not yet done D-03
Contact tree check	Quarterly	Every name in Section 4 answers; password-manager entries open	Not yet done

Drill record

Date	Drill	Measured RTO	Data loss	Gaps found
2026-09-24	Point-in-time restore (scripts /backup/pitr-drill.sh)	45 s (database only; the app restart comes on top)	None up to the target minute; the un-archived tail is at most one WAL segment (a few minutes)	(1) Postgres refused the in-progress WAL segment ("has wrong size") and aborted the whole recovery instead of stopping at the last complete record — both restore scripts now zero-pad that segment to 16 MB and fail fast when the container exits. (2) restore-from-m365.sh hard-coded the PRODUCTION volume name in its fresh-volume step (fixed: PG_VOLUME override); Section 6 said "on the dev database" — the drill actually needs the WAL archive, so it runs from /root/backups-archive into a throwaway volume

The full rebuild drill (D-01) is still to be recorded.

10 Findings and action plan

ID	Priority	Finding	Action	Effort	Owner
D-01	Now	The rebuild has never been run on a clean machine; RTO is an estimate	A clean Ubuntu 24.04 guest (KVM, 4 vCPU / 8 GB / 80 GB) is booted on the production host, isolated from it; the timed driver <code>/root/drill-vm/run-drill.sh</code> copies the RESTORE folder in and runs the kit with the passphrase. Handling the passphrase is reserved to the owner, so the owner starts it; the log records the real RTO. Record the result here and remove the guest afterwards	M	Owner (start), Engineering
D-02	Closed	Point-in-time restore untested	Drilled 2026-09-24 with <code>scripts/backup/pitr-drill.sh</code> into a throwaway container on the production host; counts verified; repeat quarterly	S	Engineering
D-03	High	No named deputies or communications lead	Fill Section 4; store the table in the password manager; phone notification on <code>it@inventre.in</code>	S	Owner
D-04	High	Database password not yet rotated (also a security item)	Run the rotation script at a quiet moment	S	Owner

ID	Priority	Finding	Action	Effort	Owner
D-05	High	Targets in Section 1 and 3 not yet approved	Owner approves or amends; the numbers then drive drill acceptance	S	Owner
D-06	High	A deliberate deletion in SharePoint could remove the backups (recycle bin only)	Step-by-step runbook written: <code>docs/runbooks/sharepoint-backup-retention-lock.md</code> (Purview retention policy on the site, 30 days, verify via the Preservation Hold library; policies do not block the agent's writes). Microsoft 365 admin applies it	S	Microsoft 365 admin
D-07	Medium	Single host; no warm standby	Written recommendation in <code>docs/runbooks/dr-decisions-and-contacts.md</code> : stay single-host with the current kit and revisit after the D-01 drill, unless the business needs RTO under 15 minutes (then a second small host with a streaming replica). Owner signs the decision	L	Owner
D-08	Closed	The maintenance page must be enabled by hand over SSH	Done 2026-09-24: <code>bash scripts/maintenance.sh on off status</code> ; runbook <code>docs/runbooks/maintenance-page.md</code> says who may flip it	S	Engineering

ID	Priority	Finding	Action	Effort	Owner
D-09	Closed	Offline-order fallback relied on a script and staff knowledge	Done 2026-09-24: docs/runbooks/offline-orders-sop.md (checklist, who decides, keying via /admin/orders/new , enrolment + phone matching, reconciliation) and the printable A4 form docs/runbooks/offline-order-form.html	S	Operations
D-10	Medium	Partner escalation contacts not centralised	Blank-field checklist of what to store per partner is in docs/runbooks/dr-decisions-and-contacts.md ; owner fills the password-manager entry	S	Owner
D-12	Medium	Orders keyed on /admin/orders/new are not sent to the Audit ERP automatically and cannot carry Magic Box selections (found while writing the offline SOP)	Either enqueue the order event from that page or keep the SOP's manual step (scripts/reemit-one-order.ts) — decide; changes an existing flow	S	Engineering
D-11	Closed	The maintenance bypass token was written in plain text in /etc/nginx/sites-available/inventre , which is world-readable (mode 644)	Done 2026-09-24: the token now lives only in /etc/nginx/inventre-maint-token.conf (mode 600, root) as nginx map variables; the site file references the variables and contains no secret; tested and reloaded	S	Engineering

Sources: backup and restore scripts under `scripts/backup/`, runbooks under `docs/runbooks/`, the HLD and LLD v1.8, Data Protection v1.2, and the operational history recorded in this engineering programme (June 2026 compromise, September 2026 outages and fixes).